LABORATORY MANUAL

Data Visualization using Tableau Public

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General Instructions for Students

- Read the Objective and Procedure of each lab exercise fully before starting the practical session.
- Follow the numbered steps in sequence; do not skip data-connection or cleaning steps, as later exercises build on them.
- Save your workbook (.twbx) after every exercise using a clear naming convention, e.g., 'RollNo_Lab3_BarPieHistogram.twbx'.
- Rename every worksheet, dashboard, and story tab meaningfully — do not leave default names such as 'Sheet 1'.
- Take a screenshot of the final output of each exercise and paste it into your practical record / lab file.
- For Lab Exercise 7, complete the base program first (7.1) before attempting the two advanced extensions (7.2 and 7.3).
- Publish at least one dashboard and one story to your Tableau Public profile before the end of the course, and note the shareable link in your lab file.
- In case of data connection errors, verify the file path, sheet name, and that Data Interpreter / correct data types have been applied.

Lab Exercise 1: Install Tableau Public Software and Examine the Workspace

Objective: To download and install Tableau Public, and to explore and understand the different components of the Tableau workspace.

Software / Data Requirements:

- Windows/macOS computer with internet access
- At least 2 GB free disk space
- Valid email ID (for Tableau Public registration)

Procedure

Part A: Installation

1. Open a web browser and navigate to https://public.tableau.com/en-us/s/download
2. Enter your email address in the sign-up box and click 'Download the App'
3. Locate the downloaded installer file (.exe for Windows / .dmg for macOS) and double-click to run it
4. Follow the on-screen setup wizard: accept the license agreement and choose the installation folder
5. Click 'Install' and wait for the installation to complete, then click 'Finish'
6. Launch Tableau Public from the Start Menu (Windows) or Applications folder (macOS)
7. Sign in with your Tableau Public account, or click 'Skip' to explore first and sign in later when publishing

Part B: Exploring the Start Page

8. Observe the 'Connect' pane on the left — this lists data source types such as Excel, Text file, JSON, and saved data sources
9. Observe the 'Open' pane in the middle — recently used workbooks are listed here
10. Observe the 'Discover' / Sample Workbooks pane on the right — Tableau Public's featured visualizations for inspiration

Part C: Navigating the Tableau Workspace

11. Connect to any sample data set (Menu: File > Open > sample workbook, or use a built-in saved data source) to load the workspace
12. Identify the Data Source page: shows tables/sheets, join/union canvas, and the data grid preview at the bottom
13. Click the 'Sheet 1' tab at the bottom to open a new worksheet
14. Identify the main components: Data pane (left, with Dimensions and Measures), Columns shelf, Rows shelf, Marks card (Color, Size, Label, Detail, Tooltip), and the view/canvas area
15. Locate the toolbar at the top: Undo/Redo, Save, New Data Source, Swap Rows/Columns, Sort, Show Me, and Presentation Mode icons
16. Click the 'Show Me' button on the top-right to view the gallery of chart types Tableau can auto-generate
17. Explore the bottom status bar: worksheet tabs, New Worksheet (+), New Dashboard, and New Story icons
18. Go to Sheet > right-click a tab > Rename Sheet to practice renaming worksheets
19. Save the workbook via File > Save As, choosing the Tableau Workbook (.twbx) format so data is packaged with the file

Expected Output

- Tableau Public is successfully installed and launches without errors
- Student can correctly identify and label the Data pane, Shelves, Marks card, Show Me panel, and toolbar
- A workbook is saved locally in .twbx format

Lab Notes / Tips

- Tableau Public workbooks can only be saved to the cloud (Tableau Public Profile) or as .twbx locally — it cannot connect to private/local data sources that cannot be embedded, unlike Tableau Desktop.
- Encourage students to hover over each icon to read its tooltip as a quick self-guided tour.

Lab Exercise 2: Connect Tableau to Data Sources, Clean Data, and Create Joins/Unions

Objective: To connect Tableau Public to Excel, CSV, and SQL data sources; clean messy data using the Data Interpreter; and combine multiple tables using joins and unions.

Software / Data Requirements:

- Sample - Superstore.xls (Excel workbook, built into Tableau)
- A CSV export of any table (e.g., Orders.csv)
- Access to a sample SQL database (MySQL/SQL Server) if available in the lab

Procedure

Part A: Connecting to an Excel Data Source

1. Open Tableau Public and click 'Microsoft Excel' under the Connect pane
2. Browse to and select the 'Sample - Superstore.xls' file, then click Open
3. On the Data Source page, observe the sheets listed on the left (Orders, People, Returns) — these act as tables
4. Drag the 'Orders' sheet onto the canvas to load it as the primary table
5. Scroll through the data grid at the bottom to preview the rows and column headers

Part B: Cleaning Data with Data Interpreter

6. Open a workbook/sheet that has extra title rows, merged cells, or subtotal rows above the actual header row
7. On the Data Source page, tick the checkbox 'Use Data Interpreter' near the top-left of the canvas
8. Click 'Review the results' to see what Tableau removed/cleaned (extra headers, blank rows, footers)
9. Verify that column headers now correctly align with the actual field names

Part C: Creating a Join

10. Drag a second table (e.g., 'People' or 'Returns') from the left pane onto the canvas next to Orders
11. Tableau automatically detects a join icon (Venn diagram) between the two tables — click it
12. In the join dialog, select the join type: Inner, Left, Right, or Full Outer
13. Set the join clause by selecting the matching key fields from each table (e.g., Region = Region)
14. Close the dialog and preview the combined data grid to confirm columns from both tables appear

Part D: Connecting to a CSV File

15. Click 'New Data Source' (the database icon with a plus) in the toolbar
16. Select 'Text file' under Connect, then browse to and open your .csv file
17. Confirm the delimiter and header row were detected correctly in the preview grid

Part E: Connecting to a SQL Data Source

18. Click 'New Data Source' > 'More…' under Connect, then choose your database (e.g., Microsoft SQL Server / MySQL)
19. Enter the Server name, Database name, and Authentication credentials, then click Sign In
20. Drag the required table(s) onto the canvas from the left-hand table list

Part F: Creating a Union

21. On the Data Source page, drag a second table with an identical structure (same column names) below the first table until the word 'Union' appears
22. Alternatively, click the drop-down on a text/Excel connection and choose 'Wildcard (Automatic) Union' to combine all CSVs in a folder matching a name pattern
23. Preview the unioned table and confirm a new 'Table Name' / 'Sheet' field is auto-added identifying the source of each row

Part G: Fixing Data Types and Renaming Fields

24. Click the data-type icon (# , Abc, calendar, etc.) above a column header to change its data type (e.g., String to Date)
25. Double-click a column header to rename a field to a more meaningful name
26. Navigate to Sheet 1 and confirm the cleaned, joined/unioned data loads correctly into the Data pane

Expected Output

- Data from Excel, CSV, and SQL sources loads correctly into Tableau
- Data Interpreter removes extraneous header/footer rows automatically
- A working join combines fields from two related tables with no orphaned rows
- A union stacks two structurally identical tables into one long table

Lab Notes / Tips

- Always check row counts before and after a join to catch accidental duplication (fan-out).
- Use 'Data > Edit Relationships' or the noodle/join icon to revisit and correct the join clause at any time.

Lab Exercise 3: Create Bar Charts, Pie Charts, and Histograms

Objective: To build and format three fundamental chart types — bar chart, pie chart, and histogram — using the Sample-Superstore data set.

Software / Data Requirements:

- Sample - Superstore data source connected (from Lab 2)

Procedure

Part A: Bar Chart

1. Open a new worksheet and connect to Sample - Superstore
2. Drag 'Category' from Dimensions to the Columns shelf
3. Drag 'Sales' from Measures to the Rows shelf
4. Click the 'Show Me' panel and confirm/select the bar chart option
5. Click the Sort icon (descending) on the toolbar to order bars by Sales
6. Drag 'Region' onto the Color mark to create a stacked/side-by-side bar comparison
7. Click 'Label' on the Marks card and check 'Show mark labels' to display data values
8. Rename the worksheet tab to 'Bar Chart – Sales by Category'

Part B: Pie Chart

9. Create a new worksheet
10. Drag 'Category' to the Color mark and 'Sales' to the Angle mark
11. In Show Me, select the pie chart icon; Tableau auto-configures the marks card to Pie
12. Drag the Size slider on the Marks card (or resize the view) to enlarge the pie
13. Drag 'Sales' again onto the Label mark; right-click the Label 'Sales' pill > Quick Table Calculation > Percent of Total to show percentage shares
14. Rename the worksheet tab to 'Pie Chart – Sales Share by Category'

Part C: Histogram

15. Create a new worksheet
16. Drag the 'Profit' measure to the Columns shelf
17. In Show Me, select the histogram icon — Tableau automatically creates a binned 'Profit (bin)' dimension and a Count of Records measure
18. Right-click the newly created 'Profit (bin)' field in the Data pane and choose 'Edit…' to change the bin size
19. Observe how the shape of the distribution changes as the bin size is adjusted
20. Rename the worksheet tab to 'Histogram – Profit Distribution'

Expected Output

- A correctly sorted, color-coded bar chart comparing Sales across Categories
- A pie chart showing each Category's percentage share of total Sales
- A histogram showing the frequency distribution of Profit values with an adjustable bin size

Lab Notes / Tips

- Pie charts are best limited to 5–6 slices; suggest a bar chart instead when categories exceed this.
- Encourage experimentation with bin sizes to discuss how bin width affects interpretation of distribution shape.

Lab Exercise 4: Create Maps and Dual-Axis Maps for Geographical Trends

Objective: To visualize data geographically using filled maps, symbol maps, and a dual-axis (layered) map combining two measures.

Software / Data Requirements:

- Sample - Superstore data source (contains State, City, Country fields)

Procedure

Part A: Basic Symbol Map

1. Open a new worksheet and connect to Sample - Superstore
2. In the Data pane, confirm 'State' has a globe icon (Geographic Role); if not, right-click State > Geographic Role > State/Province
3. Double-click the 'State' field — Tableau automatically generates Longitude and Latitude fields and plots a map
4. Drag 'Sales' onto the Color mark and 'Profit' onto the Size mark
5. Change the Marks card mark type dropdown to 'Circle' if not already selected

Part B: Filled (Choropleth) Map

6. Change the Marks card dropdown from Circle/Automatic to 'Map' (filled polygon)
7. Ensure 'Sales' is on the Color mark; observe states shaded by intensity of Sales
8. Right-click the Color legend and choose a diverging or sequential color palette suited to the data (e.g., Orange-Blue Diverging)

Part C: Building a Dual-Axis Map

9. Duplicate the Latitude pill: on the Rows shelf, hold Ctrl (Windows) / Cmd (Mac) and drag the existing Latitude pill next to itself to create a second copy
10. Right-click the second Latitude pill and select 'Dual Axis'
11. Right-click either axis and choose 'Synchronize Axis' so both layers align geographically
12. On the Marks card, click the tab for the first Latitude layer and set its mark type to 'Filled Map' with Sales on Color
13. Click the tab for the second Latitude layer and set its mark type to 'Circle' with Profit on Size and a contrasting Color
14. Add 'State' and 'Sales'/'Profit' to Tooltip on both layers for detail on hover
15. Rename the worksheet tab to 'Dual-Axis Map – Sales & Profit by State'

Expected Output

- A filled map (choropleth) showing Sales intensity by state
- A dual-axis map layering a filled Sales map beneath proportionally sized Profit circles
- Correctly synchronized axes so both map layers align geographically

Lab Notes / Tips

- If a location field shows an 'unknown' marker/exclamation icon, check for typos or ambiguous city names and resolve via 'Edit Locations'.

- Map > Background Maps lets students switch between Normal, Light, Dark, and Streets base layers.

Lab Exercise 5: Create a Hierarchy with Drill-Down Interactivity

Objective: To organize related dimensions into a hierarchy and use drill-down/drill-up interactivity to explore data at multiple levels of granularity.

Software / Data Requirements:

- Sample - Superstore data source (Category, Sub-Category and/or Country, State, City fields)

Procedure

Part A: Creating a Hierarchy

1. In the Data pane, click on 'Category' to select it, then hold Ctrl/Cmd and click 'Sub-Category' to select both
2. Right-click on the selection and choose 'Hierarchy' > 'Create Hierarchy…'
3. Name the hierarchy 'Product Hierarchy' and click OK
4. Confirm a new expandable folder-like field appears in the Data pane containing Category and Sub-Category
5. Repeat the process for 'Country', 'State', and 'City' to build a 'Geography' hierarchy

Part B: Using Drill-Down Interactivity

6. Open a new worksheet; drag the 'Product Hierarchy' field to the Rows shelf
7. Drag 'Sales' to the Columns shelf and choose a bar chart from Show Me
8. Observe that only the top level (Category) is shown initially, with a small '+' icon next to each value
9. Click the '+' icon next to a Category (e.g., Furniture) to drill down and reveal its Sub-Categories
10. Click the '-' icon to collapse and drill back up to the Category level
11. Repeat using the 'Geography' hierarchy on a map or bar chart to drill from Country → State → City
12. Rename the worksheet tab to 'Hierarchy Drill-Down – Product & Geography'

Expected Output

- A functioning Product Hierarchy (Category > Sub-Category) and Geography Hierarchy (Country > State > City)
- A worksheet that expands/collapses interactively via + and – icons to reveal deeper levels of detail

Lab Notes / Tips

- Hierarchies only work when the fields dragged are actually related (e.g., a Sub-Category always belongs to one Category).
- Students can reorder fields within a hierarchy by dragging them within the Data pane folder.

Lab Exercise 6: Build Treemaps, Heatmaps, and Scatter Plots with Trend Lines and Clustering

Objective: To construct a treemap and heatmap for categorical comparison, and a scatter plot enhanced with a trend line and k-means clustering.

Software / Data Requirements:

- Sample - Superstore data source

Procedure

Part A: Treemap

1. Open a new worksheet
2. Drag 'Category' and 'Sub-Category' to the Detail mark (or select both in Show Me directly)
3. Drag 'Sales' to the Size mark and 'Profit' to the Color mark
4. Open Show Me and select the Treemap icon to auto-arrange the marks
5. Right-click the Color legend and apply a diverging palette so loss-making sub-categories appear in red and profitable ones in green/blue

Part B: Heatmap (Highlight Table)

6. Open a new worksheet
7. Drag 'Category' to Columns and 'Region' to Rows
8. Drag 'Sales' to Color and also to Label
9. Open Show Me and select the Highlight Table / Heat Map option
10. Adjust the color palette and add borders (Format > Borders) for readability

Part C: Scatter Plot with Trend Line

11. Open a new worksheet
12. Drag 'Sales' to Columns and 'Profit' to Rows (both as continuous/aggregated measures) to create a single point
13. Drag 'Sub-Category' to the Detail mark to break the single point into multiple points, one per sub-category
14. Open the Analytics pane (left side, next to Data pane tab) and drag 'Trend Line' onto the view, dropping it on the 'Linear' option
15. Hover over the trend line to view its equation, R-squared value, and p-value in the tooltip

Part D: Adding Clustering

16. On the same scatter plot, open the Analytics pane and drag 'Cluster' onto the view
17. In the Clusters dialog, set the desired number of clusters (or leave on 'Automatic') and click outside to apply
18. Observe that Tableau adds a 'Clusters' field to the Color mark, grouping similar Sub-Categories
19. Right-click the Clusters field in the Data pane and select 'Describe Clusters' to view cluster statistics
20. Rename the worksheet tab to 'Scatter Plot – Sales vs Profit with Clustering'

Expected Output

- A treemap sized by Sales and colored by Profit for all Sub-Categories
- A heatmap (highlight table) showing Sales intensity across Category and Region
- A scatter plot of Sales vs Profit with a linear trend line and its statistical summary
- The same scatter plot segmented into clusters of similar-performing sub-categories

Lab Notes / Tips

- Clustering in Tableau uses a k-means algorithm on standardized variables — remind students that variable scale affects cluster shape.
- A trend line's p-value below 0.05 is generally considered a statistically significant relationship.

Lab Exercise 7: Create Calculated Fields and Use Parameters to Change Metrics Dynamically

Objective: To create a calculated field for Profit Ratio and build a Parameter that lets the user dynamically switch which measure is displayed in a chart.

Software / Data Requirements:

- Sample - Superstore data source

Procedure

Part A: Creating the Profit Ratio Calculated Field

1. Right-click an empty area of the Data pane and choose 'Create Calculated Field…'
2. Name the field 'Profit Ratio'
3. Enter the formula:

```
SUM([Profit]) / SUM([Sales])
```

4. Click OK; right-click the new field in the Data pane and set Default Properties > Number Format > Percentage

Part B: Creating a Parameter

5. Right-click in the Data pane and choose 'Create Parameter…'
6. Name it 'Select Metric', set Data Type to String
7. Under Allowable values, choose 'List' and add three values: Sales, Profit, Profit Ratio
8. Click OK to create the parameter

Part C: Building a Metric-Selector Calculated Field

9. Create a new calculated field named 'Metric Selector' with the formula:

```
CASE [Select Metric]   WHEN 'Sales' THEN SUM([Sales])   WHEN 'Profit' THEN SUM([Profit])
WHEN 'Profit Ratio' THEN [Profit Ratio] END
```

10. Drag 'Category' to Columns and 'Metric Selector' to Rows, and build a bar chart
11. Right-click the 'Select Metric' parameter in the Data pane and choose 'Show Parameter Control'
12. Use the parameter control on the view to switch between Sales, Profit, and Profit Ratio and observe the chart update live
13. Rename the worksheet tab to 'Dynamic Metric Selector'

Expected Output

- A Profit Ratio field formatted as a percentage
- A working parameter control that dynamically swaps the measure displayed in the bar chart

Lab Notes / Tips

- Parameters are NOT tied to a single data field — they are global workbook values referenced by calculated fields.
- This exercise has two advanced extensions immediately following (7.2 and 7.3) that build more complex calculated logic on top of this base.

Lab Exercise 7.2: Customer Value Segmentation using LOD Expressions and Nested Logic (Advanced Extension)

Objective: To classify every customer into High/Medium/Low value segments using a Fixed LOD expression, and let the user dynamically adjust the segmentation threshold through a parameter.

Complexity Level: Higher — introduces Level of Detail (LOD) expressions, nested IF/CASE logic, and a numeric parameter that reclassifies data dynamically.

Procedure

Step 1: Compute Per-Customer Total Sales with an LOD Expression

1. Create a calculated field named 'Customer Total Sales' with the formula:

```
{ FIXED [Customer Name] : SUM([Sales]) }
```

2. This LOD expression returns each customer's total sales regardless of which other dimensions are in the view — unlike a normal aggregation, it will not change if you later add Category or Region to the sheet.

Step 2: Create a Parameter for the Segmentation Threshold

3. Create a Parameter named 'High Value Threshold', Data Type Float, Allowable values 'Range', minimum 500, maximum 20000, step size 500, default 5000
4. Create a second Parameter named 'Medium Value Threshold', Data Type Float, Range 100 to 10000, step 250, default 1500

Step 3: Build a Nested Calculated Field for Segments

5. Create a calculated field named 'Customer Segment' with nested IF logic:

```
IF [Customer Total Sales] >= [High Value Threshold] THEN 'High Value' ELSEIF [Customer
Total Sales] >= [Medium Value Threshold] THEN 'Medium Value' ELSE 'Low Value' END
```

Step 4: Add a Profitability Cross-Check with a Second LOD

6. Create a calculated field named 'Customer Profit Ratio' with the formula:

```
{ FIXED [Customer Name] : SUM([Profit]) } / { FIXED [Customer Name] : SUM([Sales]) }
```

7. Create a further calculated field 'Segment Risk Flag' that flags high-value but low-margin customers:

```
IF [Customer Segment] = 'High Value' AND [Customer Profit Ratio] < 0.1 THEN 'High Value /
Low Margin – Review' ELSE 'Normal' END
```

Step 5: Visualize the Segmentation

8. Create a new worksheet; drag 'Customer Name' to Detail and 'Customer Total Sales' to Columns, 'Customer Profit Ratio' to Rows to build a scatter plot
9. Drag 'Customer Segment' to Color and 'Segment Risk Flag' to Shape
10. Show both parameter controls (right-click each Parameter > Show Parameter Control) and drag their sliders to see customers dynamically move between segments
11. Rename the worksheet tab to 'Advanced – Customer Segmentation (LOD)'

Expected Output

- A scatter plot of customers by Total Sales vs Profit Ratio, colored by dynamically computed segment
- Segment boundaries that visibly shift in real time as the threshold parameters are adjusted
- A distinct visual flag identifying high-revenue but low-margin 'at risk' customers

Lab Notes / Tips

- FIXED LOD expressions compute independently of the view's level of detail — contrast this with a plain SUM([Sales]) which would change if a dimension like Category were added to the sheet.
- Discuss the difference between FIXED, INCLUDE, and EXCLUDE LOD expressions as a follow-up conceptual question.

Lab Exercise 7.3: Parameter-Driven What-If Discount Simulation with Table Calculations (Advanced Extension)

Objective: To build a what-if simulation that lets a user test how an across-the-board discount change would affect profit over time, using parameters, calculated fields, and table calculations together.

Complexity Level: Highest — combines a what-if numeric parameter, a compound calculated field, and Quick Table Calculations (Running Total and Percent Difference) over time.

Procedure

Step 1: Create the What-If Parameter

1. Create a Parameter named 'Simulated Discount %', Data Type Float, Allowable values 'Range', minimum 0.00, maximum 0.50, step size 0.01, default 0.00, Display format Percentage

Step 2: Build the Adjusted Profit Calculated Field

2. Create a calculated field named 'Adjusted Sales' that applies the simulated discount on top of the existing discount:

```
SUM([Sales]) * (1 - [Simulated Discount %])
```

3. Create a second calculated field named 'Adjusted Profit' that recalculates profit under the simulated discount, assuming cost stays constant:

```
[Adjusted Sales] - (SUM([Sales]) - SUM([Profit]))
```

Step 3: Build a Time-Series View

4. Create a new worksheet; drag 'Order Date' to Columns and set its granularity to continuous Month (right-click the pill > select Month under the 'Continuous' section)
5. Drag 'Profit' to Rows to plot the original profit line
6. Drag 'Adjusted Profit' to Rows as a second measure to create a dual-axis line chart; right-click the second axis and choose 'Dual Axis', then 'Synchronize Axis'

Step 4: Add Table Calculations

7. Right-click the 'Adjusted Profit' pill on the Rows shelf and choose 'Quick Table Calculation' > 'Running Total' to see cumulative simulated profit over time
8. Duplicate the Adjusted Profit measure onto Rows a third time; apply Quick Table Calculation > 'Percent Difference From' > previous period, to show month-over-month change under the simulated discount
9. Add 'Rank' as a further Quick Table Calculation on a separate worksheet ranking months by Adjusted Profit, to identify the best and worst simulated months

Step 5: Make it Interactive

10. Right-click the 'Simulated Discount %' parameter and choose 'Show Parameter Control'
11. Move the parameter slider from 0% to 20% and observe both the Running Total and Percent Difference table calculations update instantly
12. Add a text object near the control explaining: 'Move the slider to simulate the effect of an additional storewide discount on monthly profit trends'
13. Rename the worksheet tab to 'Advanced – What-If Discount Simulation'

Expected Output

- A dual-axis time series comparing actual Profit against a parameter-driven Adjusted Profit
- A running total line showing cumulative simulated profit impact over the selected time period
- A percent-difference table calculation showing month-over-month volatility under the simulated discount
- A fully interactive what-if tool controlled entirely by a single parameter slider

Lab Notes / Tips

- Table calculations are computed after aggregation and depend on the fields present in the view and their sort/partition order (Compute Using) — a common source of student errors.
- This exercise combines all Lab 7 concepts (calculated fields + parameters) with table calculations, previewing the time-trend techniques used later in Lab 8.

Lab Exercise 8: Analyze Trends Over Time Using Date Fields

Objective: To build time-series visualizations that reveal sales and profit trends using Tableau's automatic date hierarchies, continuous/discrete date handling, and reference lines.

Software / Data Requirements:

- Sample - Superstore data source (Order Date field)

Procedure

Part A: Basic Trend Line

1. Open a new worksheet
2. Drag 'Order Date' to the Columns shelf — Tableau auto-creates a discrete YEAR() hierarchy
3. Drag 'Sales' to the Rows shelf
4. Open Show Me and select the line chart option

Part B: Changing Date Granularity

5. Right-click the Order Date pill on Columns and explore the 'Discrete' options: Year, Quarter, Month, Week, Day
6. Right-click again and switch to the 'Continuous' section, selecting Month, to produce a single smooth continuous trend line instead of separate per-year blocks
7. Click the '+' icon on the Order Date pill (if discrete) to drill from Year down to Quarter and Month

Part C: Comparing Multiple Trends

8. Drag 'Category' onto the Color mark to split the single trend line into three colored lines, one per category
9. Drag 'Region' onto the Detail or Filter shelf and explore filtering the view to a single region

Part D: Adding Reference Lines and Forecasting

10. Open the Analytics pane and drag 'Average Line' onto the view, dropping it at the 'Table' scope
11. Drag 'Forecast' from the Analytics pane onto the view to extend the trend line with Tableau's built-in forecasting model
12. Right-click the forecast shaded area and choose 'Forecast > Describe Forecast' to review model quality
13. Rename the worksheet tab to 'Sales Trend Over Time'

Expected Output

- A continuous monthly trend line of Sales for the full date range
- A multi-line chart comparing trends across Categories
- A trend line with an average reference line and a forecast extension

Lab Notes / Tips

- Discrete date fields create separate axis headers (good for comparing distinct periods); continuous date fields create a single connected axis (good for showing an unbroken trend).
- Forecasting requires at least a few periods of historical data and works best with regularly spaced dates.

Lab Exercise 9: Create a Dashboard with Filters and Parameter Controls

Objective: To assemble multiple worksheets into a single interactive dashboard, and add filters and parameter controls that let end-users explore the data themselves.

Software / Data Requirements:

- At least 3 completed worksheets from previous labs (e.g., Bar Chart, Map, Trend Line)

Procedure

Part A: Setting Up the Dashboard

1. Click the 'New Dashboard' icon at the bottom of the workbook (next to the worksheet tabs)
2. In the Dashboard pane on the left, set the Size to 'Automatic' or a fixed size such as 1200 x 800
3. Drag a completed worksheet (e.g., 'Bar Chart – Sales by Category') from the left pane onto the canvas
4. Drag a second worksheet (e.g., the Map from Lab 4) onto the canvas beside or below the first
5. Drag a third worksheet (e.g., the Trend Line from Lab 8) to complete the layout
6. Switch between 'Tiled' and 'Floating' layout options for individual objects via the drop-down arrow on each object

Part B: Adding Interactive Filters

7. Click on the Bar Chart within the dashboard to select it, then click the funnel ('Use as Filter') icon that appears on its top-right corner
8. Click a bar (e.g., 'Furniture') and confirm the Map and Trend Line automatically filter to show only that category
9. Drag the 'Region' field from a worksheet's Filters shelf onto the dashboard, or use Dashboard > right-click a sheet > 'Filters' > select a field to show it as a Filter Control (e.g., a drop-down or checkbox list)
10. Right-click the new filter control and choose 'Apply to Worksheets' > 'All Using This Data Source' so it filters every relevant sheet

Part C: Adding Parameter Controls

11. From the Dashboard pane's object list, drag a 'Parameter' control (e.g., the 'Select Metric' parameter from Lab 7) onto the dashboard canvas
12. Position and resize the parameter control near the relevant chart
13. Test that changing the parameter value updates the corresponding chart(s) live

Part D: Polishing the Dashboard

14. Add a Text object as a title banner at the top describing the dashboard's purpose
15. Add a Legend object if not already visible, and reposition it neatly
16. Use Dashboard > 'Device Preview' to check how the layout adapts to desktop, tablet, and phone
17. Rename the dashboard tab to 'Superstore Executive Dashboard'

Expected Output

- A single dashboard combining a bar chart, a map, and a trend line
- Cross-filtering: clicking a bar filters the map and trend line automatically
- A visible filter control (e.g., Region) and a parameter control that update the dashboard interactively

Lab Notes / Tips

- Keep dashboards to 3–5 views maximum to avoid clutter and maintain fast rendering.
- 'Use as Filter' creates action-based filtering; this differs from a Filter Control, which is a persistent on-screen widget — both should be demonstrated.

Lab Exercise 10: Create a Tableau Story to Narrate Insights

Objective: To sequence multiple worksheets and dashboards into a Tableau Story that narrates a coherent, guided data analysis for an audience.

Software / Data Requirements:

- Completed worksheets and the dashboard from Lab 9

Procedure

Part A: Starting a New Story

1. Click the 'New Story' icon at the bottom of the workbook
2. In the Story pane on the left, set the Size to match your dashboard (e.g., 1200 x 800) or choose a preset such as 'Presentation'
3. Double-click the default title text box at the top and rename it, e.g., 'Superstore Performance: A Data Story'

Part B: Adding Story Points

4. Drag the first worksheet or dashboard (e.g., the overall Sales Trend from Lab 8) onto the blank story point canvas
5. Click the caption box below the navigator and type a short insight, e.g., 'Sales have grown steadily since 2015, with a seasonal peak each November'
6. Click 'Add a new blank point' (the small box to the right of the current point in the navigator strip) to create a second story point
7. Drag a second sheet (e.g., the Bar Chart by Category) onto this new point, and edit its caption to highlight a category-level insight
8. Repeat to add a third point using the Map from Lab 4, captioning a geographic insight (e.g., which states drive the most sales)
9. Add a fourth and final point using the full Dashboard from Lab 9, captioning it as a summary/'Explore the data yourself' point

Part C: Highlighting and Annotating

10. On any story point, click a mark (e.g., a bar or map point) and use 'Highlighted' formatting under the caption's 'Update' options so the highlight is captured for that point
11. Right-click on the view and choose 'Annotate > Mark' to add a callout pointing to a specific data point, such as the top-performing state
12. Use the navigator arrows (< and >) at the top of the story to move between points and confirm each highlight/annotation is preserved per point

Part D: Finalizing and Publishing

13. Reorder story points if needed by dragging them left/right within the navigator strip
14. Preview the complete narrative by clicking through all points in Presentation Mode (press F7 or click the Presentation Mode icon)
15. Save the workbook, then use File > 'Save to Tableau Public As…' to publish the story (and dashboard) to your Tableau Public profile
16. Copy the shareable public link and confirm the story displays correctly in a browser

Expected Output

- A four-point Tableau Story that guides the viewer from an overall trend, to category detail, to geography, to an interactive summary dashboard
- Captions that clearly state the insight of each story point
- At least one annotation/highlight emphasizing a key data point
- The story successfully published and viewable on Tableau Public

Lab Notes / Tips

- Each story point independently remembers the filter/zoom/highlight state of the underlying sheet at the moment it was captured — remind students to finalize a point's view before moving to the next.
- A good story follows a narrative arc: Context → Insight → Insight → Call-to-action/Exploration.